Integrales impropias. Sea $f \in \mathcal{R}$ en $[a,t]$ $\forall t > a$.

$$\int_a^\infty f(x)\,dx \overset{def}{=} \lim_{t\to\infty} \int_a^t f(x)\,dx.$$

Si $f(x)$ tiene antiderivada $F(x)$ en $[a,\infty)$

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty} F(t) - F(a).$$

Problema: $\displaystyle\int_0^x \frac{dt}{2+\cos^2 t}$.

Sln:

$$\frac{dt}{2+\cos^2 t} = \frac{dt}{3\cos^2 t + 2\operatorname{sen}^2 t} = \frac{dt}{\cos^2 t}\cdot\frac{1}{(3+2\tan^2 t)}$$

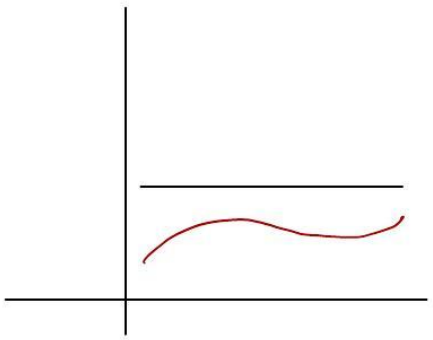
$$d(\tan t) = \frac{dt}{\cos^2 t} \qquad \text{Sea } u = \tan t \qquad v = \sqrt{\tfrac{2}{3}}\,u.$$

$$= \frac{du}{3\left(1+\frac{2}{3}u^2\right)} = \frac{\sqrt{3}}{3\sqrt{2}}\,\frac{dv}{1+v^2} = \frac{1}{\sqrt{6}}\,\frac{dv}{1+v^2}$$

$$F(x) = \int_0^x \frac{dt}{2+\cos^2 t} = \frac{1}{\sqrt{6}} \int_0^{\sqrt{\frac{2}{3}}\tan x} \frac{dv}{1+v^2} = \frac{1}{\sqrt{6}} \arctan\left(\sqrt{\tfrac{2}{3}}\tan x\right)$$

$f(t) = \dfrac{1}{2+\cos^2 t}$ $\quad f(t)$ es acotada. $\quad |f(t)| < \frac{1}{3}$.

$|F(x)| < \frac{1}{\sqrt{6}}\,\frac{\pi}{2}$, i.e, $F(x) \geq \frac{1}{3}x$. luego.



$A > a$.

$\displaystyle\int_a^\infty f(x)\,dx$ converge $\iff \int_A^\infty f(x)\,dx$ converge

$\displaystyle\int_a^\infty f(x)\,dx$ converge $\iff \int_a^\infty c\,f(x)\,dx$ converge

$\displaystyle\int_a^\infty f(x)\,dx,\ \int_a^\infty g(x)\,dx < \infty \Rightarrow \int_a^\infty (f(x) \pm g(x))\,dx < \infty.$

Sea $f(x) \geq 0$

$$\varphi(t) = \int_a^t f(x)\,dx \quad \text{es creciente.}$$

$\exists \lim_{t\to\infty} \varphi(t) \iff \varphi(t)$ es acotada.

$f(x) \geq 0$ y $g(x) \geq 0$.

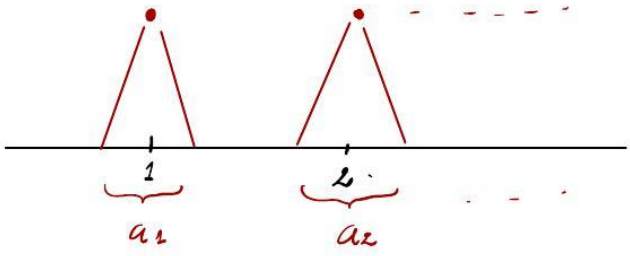
i) $\left.\begin{array}{l} \int_a^\infty f(t)\,dt < \infty \\ g(t) < f(t) \end{array}\right\} \Rightarrow \int_a^\infty g(t)\,dt < \infty.$

ii) $\lim_{x\to\infty} \frac{f(x)}{g(x)} = K \neq 0$ entonces $\int_a^\infty f(t)\,dt$, $\int_a^\infty g(t)\,dt$ convergen simultaneamente.

Problema 1. Sea f continua en $[0,\infty)$.

$$\int_0^\infty f(x)\,dx \text{ converge} \Rightarrow \lim_{x\to\infty} f(x) = 0. \quad \{\text{Verdad o falso}\}.$$

Falso: Sea $\sum_{n=1}^{\infty} a_n$ una serie convergente de términos positivos



Son triángulos de base a_n y las alturas igual a $\frac{1}{2}$.

Sea A_n el área del triángulo con base A_n.

$$\int_0^\infty f(x)\,dx = \sum_{n=1}^{\infty} A_n = \sum_{n=1}^{\infty} \frac{a_n}{4} < \infty \quad \text{Sin embargo} \quad \lim_{x\to\infty} f(x) \neq 0.$$

Problema 2. Presentar ejemplo de $f(x)$ continua, no acotada t.q $\int_0^\infty f(x)\,dx$ converge y otro que converge $\int_0^\infty f(x)\,dx = 0$.

Integrales de función no acotada

$f(x)$ integrable en $[a,t]$ $\forall t < b$

Thm. $\int_a^\infty f(x)\,dx < \infty \iff \forall \varepsilon > 0\ \exists \delta > 0$ t.q $A_1 > A_2 > \delta \Rightarrow \left| \int_{A_1}^{A_2} f(x)\,dx \right| < \varepsilon.$

Thm. $\int_a^\infty |f(x)|\,dx$ converge entonces $\int_a^\infty f(x)\,dx$ converge.

Proof. (Para usted).

Thm (Valor medio) $f(x)\geq 0$ decreciente en $[a,b]$ $g(x)$ es integrable en $[a,b]$ entonces $\exists\, \xi\in[a,b]$ t.q

$$I=\int_a^b f(x)g(x)\,dx = f(a)\int_a^\xi g(x)\,dx.$$

Proof. Sea $\{x_0,\dots,x_n\}$-particion de $[a,b]$.

$$I=\sum_{i=0}^{n-1}{}'\int_{x_i}^{x_{i+1}} f(x)g(x)\,dx = \sigma+\rho.$$

$$\sigma=\sum_{i=0}^{n-1}{}' f(x_i)\int_{x_i}^{x_{i+1}} g(x)\,dx \qquad \rho=\sum_{i=0}^{n-1}{}'\int_{x_i}^{x_{i+1}} (f(x)-f(x_i))\,g(x_i)$$

$|g(x)|<\mathcal{L}$ en $[a,b]$.

$$|f(x)-f(x_i)|\leq M_i-m_i \qquad M_i=\sup_{x\in\Delta x_i} f(x) \qquad m_i=\inf_{x\in\Delta x_i} f(x)$$

$$|\rho|\leq \mathcal{L}\sum_{i=0}^{n-1}{}'(M_i-m_i)\,\Delta x_i \Rightarrow \lim_{|P|\to 0}\rho=0.$$

$G(x)=\int_a^x g(t)\,dt$ - continua en $[a,b]$.

$$\sigma=\sum_{i=0}^{n-1}{}' f(x_i)\big(G(x_{i+1})-G(x_i)\big)$$

$$=\sum_{i=1}^{n}{}' f(x_{i-1})G(x_{i+1})-\sum_{i=0}^{n-1}{}' f(x_i)G(x_i)$$

$$=\sum_{i=1}^{n-1}{}'\big(f(x_{i-1})-f(x_i)\big)G(x_i)+f(x_{n-1})G(x_n)$$

($f(x_{i-1})-f(x_i)\geq 0$; $f(x_{n-1})\geq 0$)

$$M=\sup_{x\in[a,b]} G(x) \qquad m=\inf_{x\in[a,b]} G(x)$$

$$\sigma\leq M\Big(\sum_{i=1}^{n-1}{}'\big(f(x_{i-1})-f(x_i)+f(x_{n-1})\big)=Mf(a).$$

$\sigma \geq m f(a)$

luego $\quad m f(a) \leq I \leq M f(a)$

$\exists \mu \in [m, M]$ t.q $\quad I = \mu f(a)$

$\exists \xi \in [a,b]$ t.q $\quad \mu = G(\xi)$.

Thm: Sea f monotona en $[a,\infty)$ y acotada si $\int_a^\infty g(x)dx < \infty$ entonces

$$\int_a^\infty f(x)g(x)dx < \infty.$$

Proof: (Criterio Abel).

$$\left| \int_{A_1}^{A_2} f(x)g(x)dx \right| = f(A_1)\int_{A_1}^{\xi} g(x)dx + f(A_2)\int_{\xi}^{A_2} g(x)dx$$

$$\leq 2L\varepsilon \quad \text{donde } |f(x)| < L$$

$\exists \delta$ t.q $\quad a_2 > a_1 > \delta$ implica $\left| \int_{a_1}^{a_2} g(x)dx \right| < \varepsilon$. ∎

Hemos usado que:

Si f es creciente y no negativa en $[a,b]$ y $g(x)$ es integrable en $[a,b] \Rightarrow \exists \xi \in [a,b]$ t.q.

$$I = \int_a^b f(x)g(x)dx = f(b)\int_{\xi}^{b} g(x)dx.$$

y además

$$\int_a^b f(x)g(x)dx = f(a)\int_a^{\xi} g(x)dx + f(b)\int_{\xi}^{b} g(x)dx.$$

Criterio Dirichlet: Sea f monotona $\lim_{x\to\infty} f(x) = 0$, $g(x)$ integrable en $[a,b]$ y $\left|\int_a^b g(x)dx\right| < K \ \forall b$. Entonces

$$\int_a^\infty f(x)g(x)dx < \infty.$$

Proof (Para usted).

Def: $$\int_{-\infty}^{\infty} f(x)\,dx \overset{def}{=} \lim_{\substack{A\to+\infty \\ B\to-\infty}} \int_B^A f(x)\,dx = \lim_{A\to\infty} \int_a^A f(x)\,dx + \int_B^a f(x)\,dx$$

$$\neq \lim_{A\to+\infty} \int_{-A}^{A} f(x)\,dx \overset{def}{=} V.p \int_{-\infty}^{+\infty} f(x)\,dx.$$

$\forall f$ impar $\quad V.p \int_{-\infty}^{\infty} f(x)\,dx = 0$ sin embargo $\nexists \int_{-\infty}^{\infty} x\,dx$.

Problema: Si $\int_{-\infty}^{+\infty} f(x)\,dx = A < \infty$ entonces $V.p \int_{-\infty}^{+\infty} f(x)\,dx = A$.